

Data Management Technologies
Unit 10 Consolidation task

Activity 1.: Administering a Database

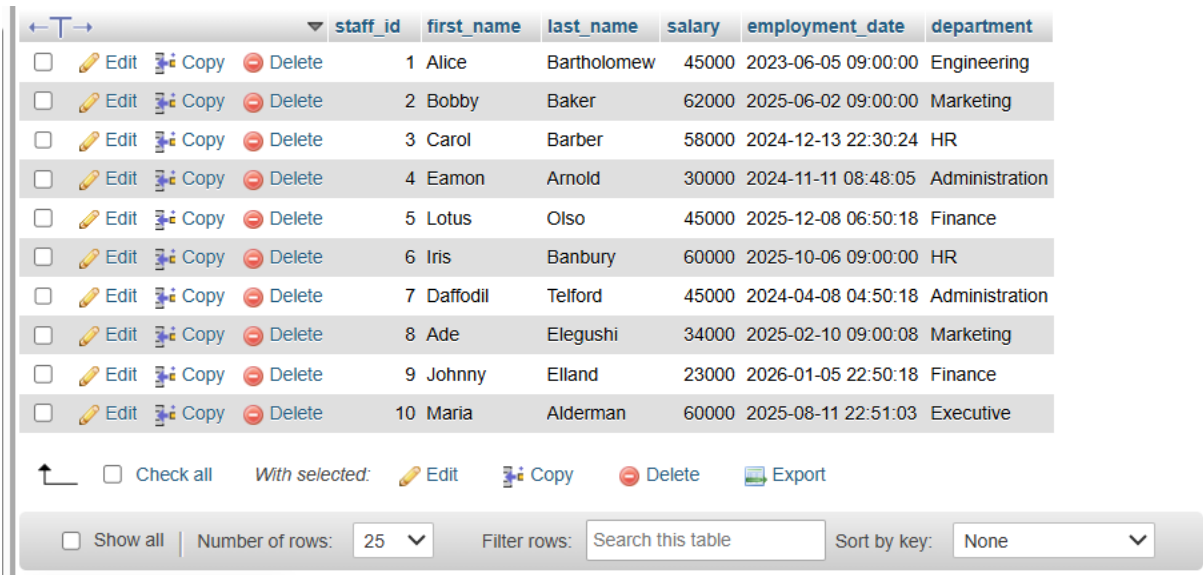
Task 1: Database Initialization

Create a simple database in MySQL or MongoDB.
Define appropriate users and assign roles (e.g., admin, read-only, data entry).
You should supply screenshots showing database creation and role assignments.

Response

Overview of database created

This is the database for a small company with details of 10 staff, including names, employment dates and pay details.



	staff_id	first_name	last_name	salary	employment_date	department
☐ Edit Copy Delete	1	Alice	Bartholomew	45000	2023-06-05 09:00:00	Engineering
☐ Edit Copy Delete	2	Bobby	Baker	62000	2025-06-02 09:00:00	Marketing
☐ Edit Copy Delete	3	Carol	Barber	58000	2024-12-13 22:30:24	HR
☐ Edit Copy Delete	4	Eamon	Arnold	30000	2024-11-11 08:48:05	Administration
☐ Edit Copy Delete	5	Lotus	Olso	45000	2025-12-08 06:50:18	Finance
☐ Edit Copy Delete	6	Iris	Banbury	60000	2025-10-06 09:00:00	HR
☐ Edit Copy Delete	7	Daffodil	Telford	45000	2024-04-08 04:50:18	Administration
☐ Edit Copy Delete	8	Ade	Elegushi	34000	2025-02-10 09:00:08	Marketing
☐ Edit Copy Delete	9	Johnny	Eiland	23000	2026-01-05 22:50:18	Finance
☐ Edit Copy Delete	10	Maria	Alderman	60000	2025-08-11 22:51:03	Executive

☐ Check all With selected: Edit Copy Delete Export

☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Image 1: Main table 'staff'

Admin users have full access (including access to DDL and DML). Can create and remove tables in SQL, using CREATE, DROP and DELETE. This is granted to administrative staff.

Read-only users have access to the SELECT keyword only, so that they can view, but not edit the data. This is granted to managers, executive staff and auditors, who require the information, but have no need to change it.

Data entry users have access to SELECT, INSERT and UPDATE keywords, so can edit the data. This includes HR staff, clerks, and receptionists who require moderate permissions to change the data.

The SQL syntax for role assignments is as below:

```
1 -- Create users
2 CREATE USER IF NOT EXISTS 'Daffodil_Telford_Admin'@'localhost' IDENTIFIED BY 'AdminPass123!';
3 CREATE USER IF NOT EXISTS 'Maria_Alderman_Executive'@'localhost' IDENTIFIED BY 'ReadOnly123!';
4 CREATE USER IF NOT EXISTS 'Carol_Barber_HR'@'localhost' IDENTIFIED BY 'DataEntry123!';
5
6 -- Assign roles/privileges
7
8 -- Admin: full control
9 GRANT ALL PRIVILEGES ON unit_10-consolidation_db.* TO 'admin_user'@'localhost';
10
11 -- Read-only: SELECT only
12 GRANT SELECT ON unit_10-consolidation_db.* TO 'readonly_user'@'localhost';
13
14 -- Data entry: INSERT and UPDATE only (no DELETE, no schema changes)
15 GRANT SELECT, INSERT, UPDATE ON unit_10-consolidation_db.* TO 'dataentry_user'@'localhost';
16
17 FLUSH PRIVILEGES;
18
19 -- Print permission information
20
21 SHOW GRANTS FOR 'admin_user'@'localhost';
22 SHOW GRANTS FOR 'readonly_user'@'localhost';
23 SHOW GRANTS FOR 'dataentry_user'@'localhost';
```

Image 2: Role Assignments

Task 2: Backups and Restorations:

Use tools (e.g., *mysqldump* for MySQL or *mongodump* for MongoDB) to back up a sample database. Simulate data loss by deleting a table or collection and restore the database from the backup. Document: backup and restoration commands, outputs, and a reflection on the process.

Response:

Simulation of data loss: The steps and screenshots are recorded below:

Back up on command prompt:

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\uchec>cd "C:\wamp64\bin\mysql\mysql8.4.7\bin"

C:\wamp64\bin\mysql\mysql8.4.7\bin>mysqldump -u root -p unit_10-consolidation > C:\Users\uchec\unit_10-consolidation_backup.sql
Enter password:

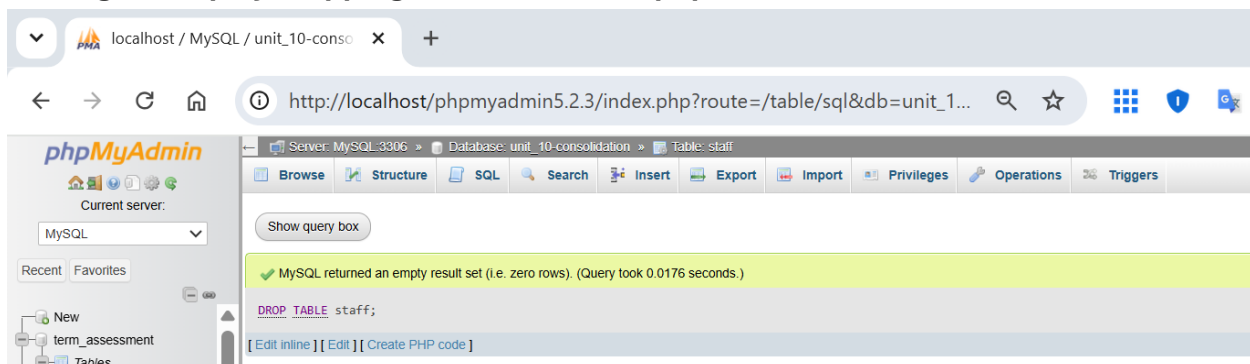
C:\wamp64\bin\mysql\mysql8.4.7\bin>dir C:\Users\uchec\unit_10-consolidation_backup.sql
Volume in drive C has no label.
Volume Serial Number is 1EA9-AA02

Directory of C:\Users\uchec

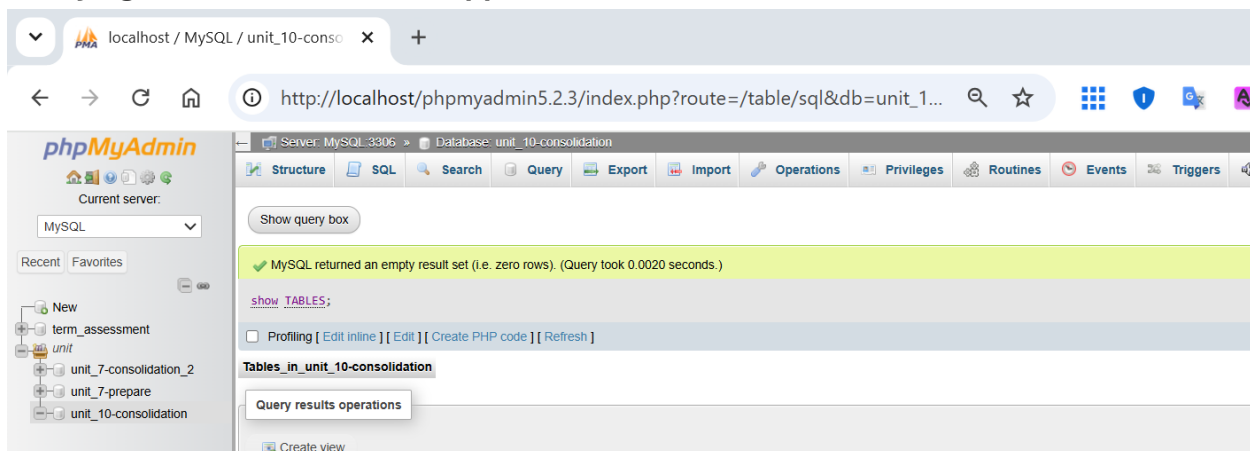
08/06/2026  23:51                2,838 unit_10-consolidation_backup.sql
             1 File(s)                2,838 bytes
             0 Dir(s) 146,493,988,864 bytes free

C:\wamp64\bin\mysql\mysql8.4.7\bin>
```

Testing backup by dropping the table staff in phpMYAdmin:



Verifying the table has been dropped:



Restoring data via command prompt

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\uchec>cd "C:\wamp64\bin\mysql\mysql8.4.7\bin"

C:\wamp64\bin\mysql\mysql8.4.7\bin>mysql -u root -p unit_10-consolidation < C:\Users\uchec\unit_10-consolidation_backup.sql
Enter password:

C:\wamp64\bin\mysql\mysql8.4.7\bin>
```

Viewing the restored data:

localhost / MySQL / unit_10-consolidation

http://localhost/phpmyadmin5.2.3/index.php?route=/sql&db=unit_10-consolidation

Server: MySQL:3306 Database: unit_10-consolidation Table: staff

Showing rows 0 - 9 (10 total, Query took 0.0022 seconds.)

`SELECT * FROM `staff``

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	staff_id	first_name	last_name	salary	employment_date	department
☐	1	Alice	Bartholomew	45000	2023-06-05 09:00:00	Engineering
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Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Task 3: Performance Monitoring

Use built-in tools (e.g., MySQL Workbench Performance Dashboard or MongoDB Atlas metrics) to monitor database performance.

Identify a scenario where the database might experience bottlenecks (e.g., high query load or large dataset).

Database performance measured via phpMyAdmin Status:

The screenshot shows the phpMyAdmin interface for a MySQL server. The 'Status' tab is selected, displaying network traffic and connection statistics. The left sidebar shows a database tree with 'unit_10-consolidation' selected. The main content area shows the following data:

Network traffic since startup: 29.6 MiB
This MySQL server has been running for 2 days, 22 hours, 39 minutes and 33 seconds. It started up on Jun 06, 2026 at 12:41 AM.

Traffic	#	per hour	Connections	#	per hour	%
Received	8.5 MiB	122.6 KiB	Max. concurrent connections	4	---	---
Sent	21.1 MiB	306.4 KiB	Failed attempts	0	0	0%
Total	29.6 MiB	429.0 KiB	Aborted	0	0	0%
			Total	10 k	142.02	100.00%

Scenario where database might experience bottlenecks

The database is a staff data management system for a medium-sized enterprise. The database might experience bottlenecks where there are multiple simultaneous connections, and too many users are logged in and trying to access data at the same time.

Bottlenecks might also occur where there are a large number of complex queries, such as payroll computing pay summaries for many individuals simultaneously.